

Firm Dynamics and Productivity Growth

Session 4: Resource Allocation and Aggregate Productivity

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A Primer on Misallocation

A Primer on Misallocation

Big question: Why are some countries rich and others poor?

$$Y = AK^{\alpha}L^{1-\alpha} \quad \Rightarrow \quad \frac{Y}{L} = A\left(\frac{K}{L}\right)^{\alpha}$$

- Cross-country differences in living standards too large to be explained by inputs
- The rest is differences in productivity

A Primer on Misallocation

- What explains differences in productivity?
 - Technology?
 - Allocation of inputs across different producers?
- **Misallocation:** productive firms “too small” and unproductive firms “too large”

A Primer on Misallocation

- In an efficient economy, firms will borrow up to the point where the marginal product of capital equals the interest rate.
- Suppose there are two identical firms: one has access to subsidized credit and the other can only borrow at a high interest rate. Then the marginal product will be lower for the firm with subsidized credit.
- The productivity loss comes from the “wrong” firms using too many resources. If capital was reallocated from the firm with low marginal product to the firm with high marginal product then output would increase.

A Primer on Misallocation

- **Main idea:** Look for *dispersion* in marginal product as evidence of *misallocation*
- In practice, marginal products are not typically observable, so measuring their dispersion requires imposing some structure

Measuring Misallocation

MISALLOCATION AND MANUFACTURING TFP IN CHINA AND INDIA*

CHANG-TAI HSIEH AND PETER J. KLENOW

Resource misallocation can lower aggregate total factor productivity (TFP). We use microdata on manufacturing establishments to quantify the potential extent of misallocation in China and India versus the United States. We measure sizable gaps in marginal products of labor and capital across plants within narrowly defined industries in China and India compared with the United States. When capital and labor are hypothetically reallocated to equalize marginal products to the extent observed in the United States, we calculate manufacturing TFP gains of 30%–50% in China and 40%–60% in India.

Hsieh–Klenow Framework

- Consider an industry s with heterogeneous firms i
- Industry output is a CES aggregate of firm varieties:

$$Y_s = \left(\sum_i Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad \sigma > 1$$

- Each firm produces using capital and labor:

$$Y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}$$

- Firms differ in physical productivity A_{si}

Firm problem

- Firm i in industry s chooses capital and labor to maximize profits:

$$\pi_{si} = (1 - \tau_{si}^Y) P_{si} Y_{si} - w L_{si} - (1 + \tau_{si}^K) R K_{si}$$

- Optimal input choices satisfy:

$$\alpha_s \frac{P_{si} Y_{si}}{K_{si}} = \left(\frac{\sigma}{\sigma - 1} \right) \left(\frac{1 + \tau_{si}^K}{1 - \tau_{si}^Y} \right) R$$

$$(1 - \alpha_s) \frac{P_{si} Y_{si}}{L_{si}} = \left(\frac{\sigma}{\sigma - 1} \right) \left(\frac{1}{1 - \tau_{si}^Y} \right) w$$

- τ_{si}^Y distorts the scale of the firm, τ_{si}^K distorts the capital-labor ratio

TFPQ and TFPR

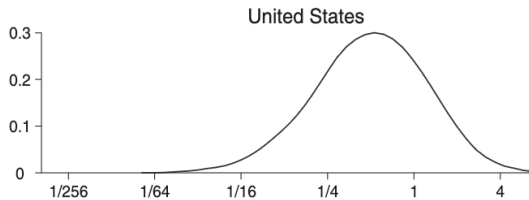
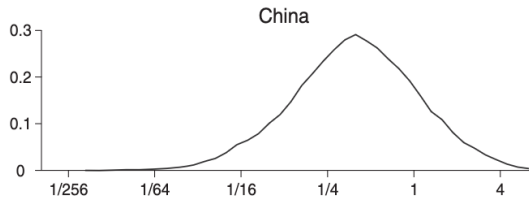
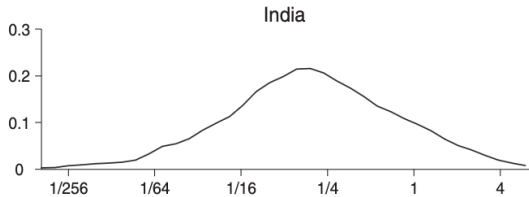
- Hsieh–Klenow distinguish between physical productivity (TFPQ) and revenue productivity (TFPR)

$$\text{TFPQ}_{si} = A_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}} \quad \text{TFPR}_{si} = P_{si} A_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}}$$

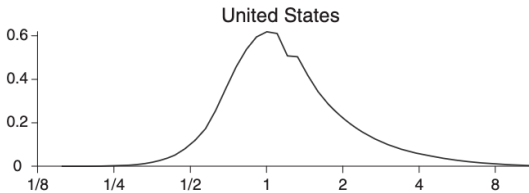
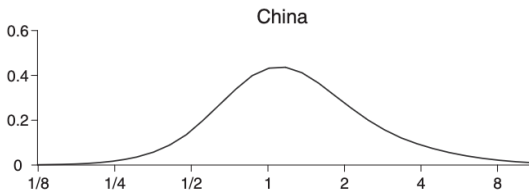
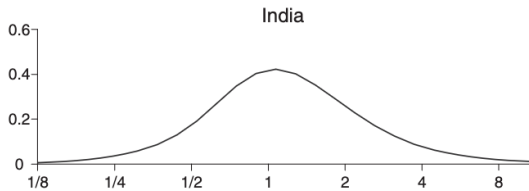
- The gap between TFP_s and TFP_s^* measures the cost of misallocation:

$$\text{TFP}_s = \left[\sum_i \left(A_{si} \frac{\overline{\text{TFPR}}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}} \quad \text{TFP}_s^* = \left[\sum_i A_{si}^{\sigma-1} \right]^{\frac{1}{\sigma-1}}$$

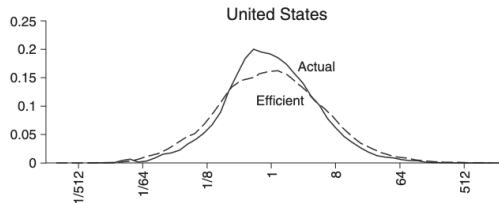
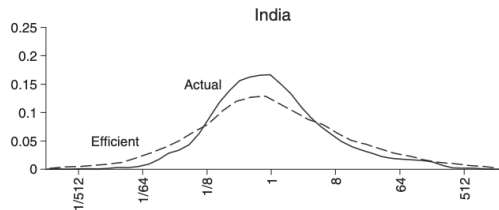
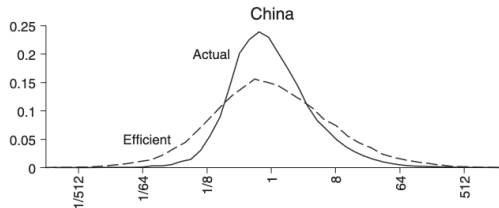
TFPQ



TFPR



Counterfactual



Dynamics of Misallocation

Dynamics of Misallocation

CAPITAL ALLOCATION AND PRODUCTIVITY IN SOUTH EUROPE*

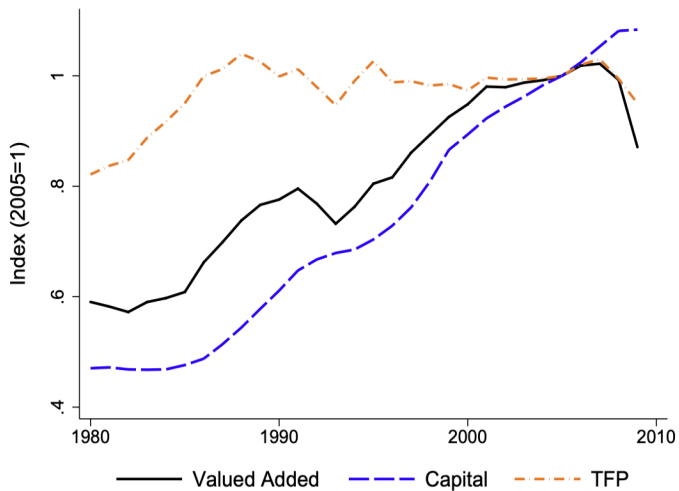
GITA GOPINATH
ŞEBNEM KALEMLI-ÖZCAN
LOUKAS KARABARBOUNIS
CAROLINA VILLEGAS-SANCHEZ

Starting in the early 1990s, countries in southern Europe experienced low productivity growth alongside declining real interest rates. We use data for manufacturing firms in Spain between 1999 and 2012 to document a significant increase in the dispersion of the return to capital across firms, a stable dispersion of the return to labor, and a significant increase in productivity losses from capital misallocation over time. We develop a model with size-dependent financial frictions that is consistent with important aspects of firms' behavior in production and balance sheet data. We illustrate how the decline in the real interest rate, often attributed to the euro convergence process, leads to a significant decline in sectoral total factor productivity as capital inflows are misallocated toward firms that have higher net worth but are not necessarily more productive. We show that similar trends in dispersion and productivity losses are observed in Italy and Portugal but not in Germany, France, and Norway. *JEL Codes:* D24, E22, F41, O16, O47.

Background

- South Europe since 1990s characterized by:
 - Declining interest rates
 - Large capital inflows
 - Low TFP growth rates
- **Hypothesis:** Low TFP because capital was misallocated after interest rate decline

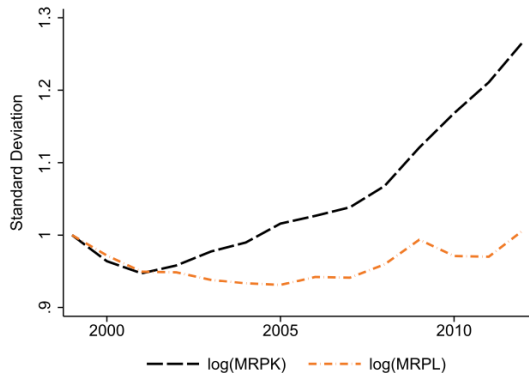
Spanish Manufacturing



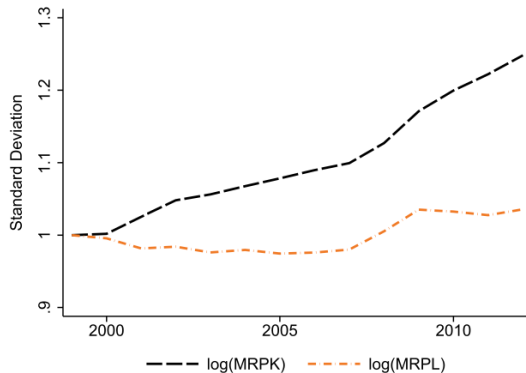
Overview

- Document **empirical** evidence of misallocation in manufacturing firms:
 - No trend in dispersion of $\log(\text{MRPL}) \rightarrow$ labor no more misallocated than before
 - Increase in dispersion of $\log(\text{MPRK}) \rightarrow$ increased capital misallocation
- Develop **model** of heterogeneous producers with financial frictions:
 - Size-dependent borrowing constraint: Large firms can borrow more
 - Fall in interest rate $\rightarrow$ TFP decline
 - How? Lower interest rate $\rightarrow$ Increases optimal capital
 - Large firms can borrow, so they increase capital (return to capital falls)
 - Small firms cannot borrow, so they cannot expand (return to capital increases)

Dispersion in MP



(a) Permanent Sample



(b) Full Sample

Model

- Firms produce varieties of manufacturing goods with production function $y = Z_{it} k_{it}^{\alpha} l_{it}^{1-\alpha}$ and face downward sloping demand $y = p_{it}^{-\epsilon}$
- Firms choose consumption, debt, investment, labor, and price to maximize $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it})$
- Productivity Z_{it} has an aggregate effect, an idiosyncratic permanent effect, and an idiosyncratic transitory effect
- Size-dependent borrowing constraint
- Costly capital adjustments

Model

Let $a_{it} = k_{it} - b_{it}$:

$$V(a, k, z^P, z^T, X) = \max_{a', k', l, p} u(c) + \beta \mathbb{E} V(a', k', z^P, z^{T'}, X')$$

$$\text{s.t.:} \quad c + a' + \frac{\psi(k' - k)^2}{2k} = p(y)y - wl - (r + \delta)k + (1 + r)a$$

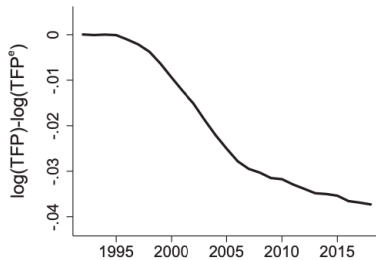
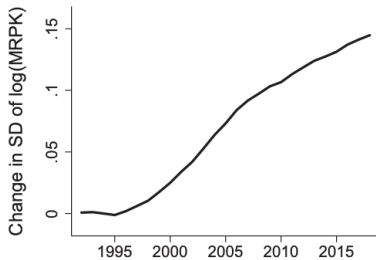
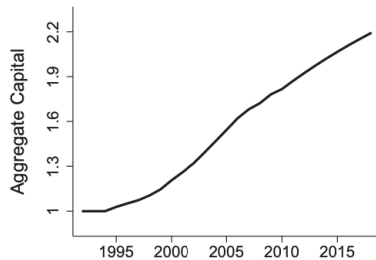
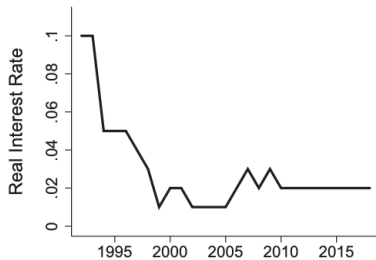
$$y = p^{-\epsilon}$$

$$y = Zk^\alpha l^{1-\alpha}$$

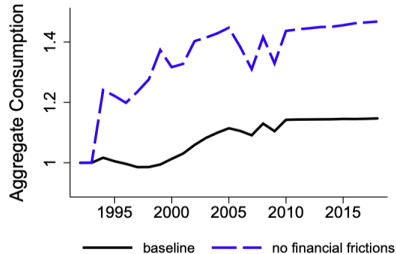
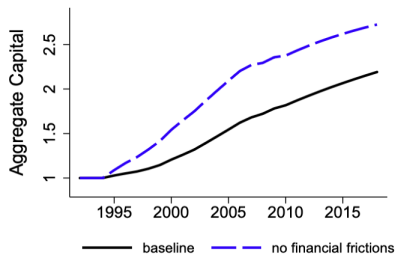
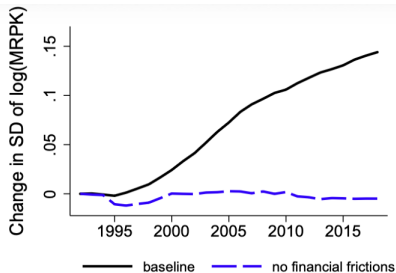
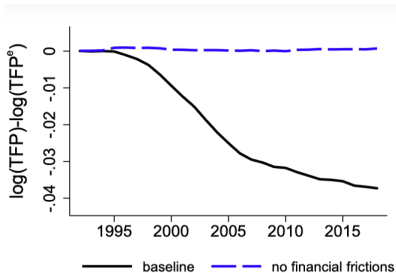
$$Z = Z^A z^P \exp(z^T)$$

$$k' \leq (\lambda_0 + \lambda_1 \Psi(k')/a') a'$$

Aggregate Variables



Financial Frictions



Mechanism

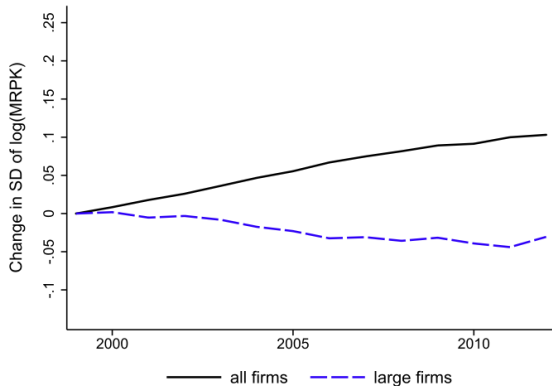
$$\frac{k_{is,07} - k_{is,99}}{k_{is,99}} = d_s + \beta_z \log(Z_{is,99}) + \beta_a \log(a_{is,99}) + \beta_k \log(k_{is,99}) + u_{is}^k$$

	Data	Model HeF	Model HoF	Model NoF
$\hat{\beta}_a$	0.17	0.21	0.11	-0.03

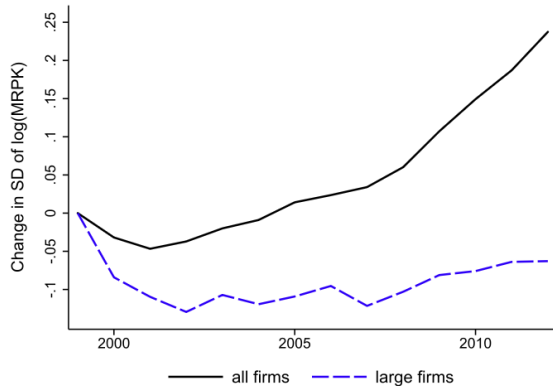
$\hat{\beta}_a = 0.17\%$: firms that in 1999 had a 1% higher net worth invested 0.17% more in capital by 2007

Why? Larger firms more likely than small firms to overcome their borrowing constraints

Mechanism

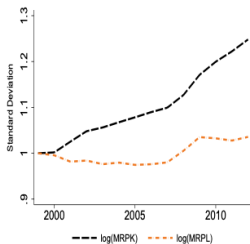


(a) Model

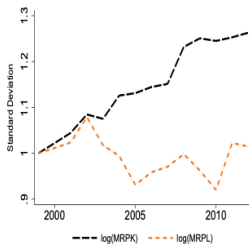


(b) Data

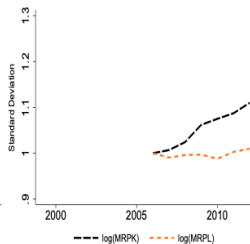
Other Countries



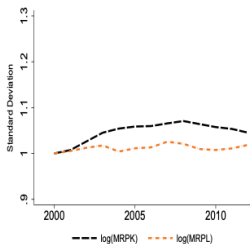
(a) Spain



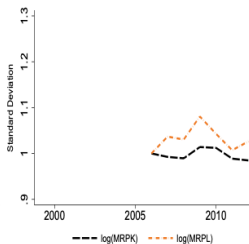
(b) Italy



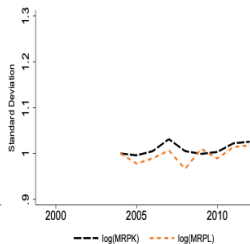
(c) Portugal



(d) France

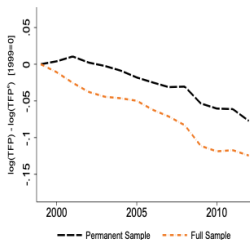


(e) Germany

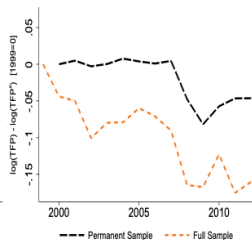


(f) Norway

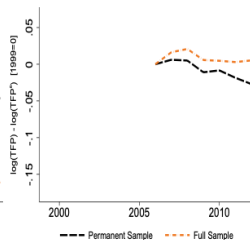
Other Countries



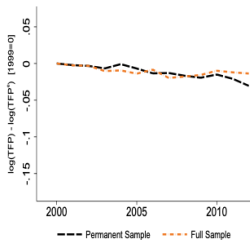
(a) Spain



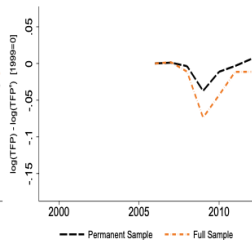
(b) Italy



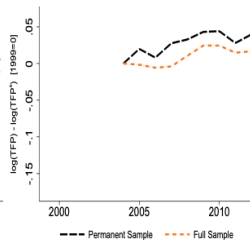
(c) Portugal



(d) France



(e) Germany



(f) Norway

Where to go from here

THE GLOBAL CAPITAL ALLOCATION PROJECT

RESEARCH DATA HUB TEACHING EVENTS PRESS ABOUT US

TEACHING AND LEARNING RESOURCES

BIG-DATA INITIATIVE IN INTERNATIONAL MACRO-FINANCE

Large-scale data-driven research has revolutionized many fields in the social sciences, but it often presents specific challenges for new researchers entering the field. This initiative seeks to lower these barriers to entry by sharing knowledge, codes, and data as a starter kit for new researchers. Each year, we bring together 30 PhD students from schools worldwide for a two-day workshop on big data research in international macro-finance.

[BIG-DATA INITIATIVE MATERIALS](#)

INTERNATIONAL MACROECONOMICS AND FINANCE PHD COURSE

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[PHD COURSE MATERIALS](#)

STEG-PEDL Virtual Course on 'Private Enterprise, Productivity, and Economic Growth'

Series started 6 February 2026 and ended 15 May 2026

WEBSITE

[REGISTER TO JOIN ONLINE](#)

Course Description: Perhaps the central question of economic development and structural transformation is how to increase productivity. As economies move out of agriculture, this question naturally hinges on the major private-sector producers in the modern economy: firms. The growing body of research on firms in lower-income countries is notable in the wide variety of both analytical methods and sources of data that are employed either separately or together. Research combines

Where to go from here

- **Measurement**

Carrillo et al. (2023); Kehrig–Vincent (2025); Bergquist–Lashkari–Verhoogen (2026)

- **Microfoundations of misallocation**

David–Venkateswaran (2019); Liang (2023); Ropele–Gorodnichenko–Coibion (2024); Gupta (2024); Bornstein–Peter (2025); Li–Lian–Ma–Martell (2025); Baqaee–Burstein (2025); Holguín–Ocampo–Salgado (2026)

- **Discrimination and inequality**

Hsieh–Hurst–Jones–Klenow (2019); Morazzoni–Sy (2022); Goraya (2023); Tan–Zeida (2024); Hsieh–Klenow–Manu–Rockall (2026)

Where to go from here

- **Misallocation of innovation**

Ottonello–Winberry (2024); Dou–Ji–Tian–Wang (2024); Aghion et al. (2025); Lehr (2025)

- **International aspects**

Bau–Matray (2023); Bai–Jin–Lu (2024); Kilumelume et al. (2025)

- **Organizational aspects**

Chauvin–Chauvin–Chaves (2025); Schymik (2025)